

Foundation Models for STT Classification

Carlo Peron

February 6, 2026

Abstract

Your abstract.

1 Introduction

Soft tissue tumors form a heterogeneous group of lesions that arise from mesenchymal tissues and present substantial diagnostic and therapeutic challenges [Fle18]. Accurate non-invasive diagnosis is essential for treatment planning and prognosis, but tumors often exhibit similar characteristics on scans, and definitive confirmation frequently requires invasive testing that may not be possible before therapy. Medical imaging advancements in combination with machine learning technology enable better diagnostic results, but their clinical application faces three main barriers: insufficient training data, high processing requirements for 3D medical images and the need for universal features that maintain stability [L⁺23, L⁺17]. The training of foundation models on extensive medical volume datasets presents a potential solution to these problems because they generate adaptable embeddings which need less labeling information while reducing training and inference expenses and enhancing performance between different healthcare facilities [W⁺23, L⁺23].

The proposed pipeline leverages large-scale pre-trained volumetric representations together with compact, data-efficient classifiers. The approach reuses semantic features learned by a foundation model for 3D medical imaging to reduce the amount of task-specific labeled data required, while a lesion-centric preprocessing strategy and region-of-interest pooling concentrate computational effort on the most informative parts of the scan. The extracted embeddings are standardized and optionally reduced in dimensionality. They are then provided to transformer-based tabular classifiers [HMEH23, T⁺24], which are optimized for small structured feature sets and enable rapid model testing without full volumetric training.

The implemented system employs SAM-Med3D [W⁺23] as a feature extractor to obtain rich embeddings from regions containing lesion, applies

adaptive ROI pooling and lesion filtering to produce compact feature vectors per study, and uses TabPFN [HMEH23] and LoCalPFN [T⁺24] as classifiers. Complementary experiments compare these transfer-learning workflows to native 3D classification heads trained on cropped volumes, allowing evaluation of the trade-offs between feature reuse and end-to-end learning under matched preprocessing conditions.

Experimental data comprise 930 CT and MRI studies from the publicly available WORC dataset [S⁺21], spanning multiple tumor categories including gastrointestinal stromal tumors, lipomatous tumors, desmoid-type fibromatosis, liver lesions, and melanoma metastases.

The research evaluates whether pre-trained 3D embeddings, combined with transformer-based tabular classifiers and limited labeled data, can achieve competitive performance on medical classification tasks. It further compares alternative data-preparation strategies for volumetric lesions and develops a systematic framework for running experiments across multiple datasets and aggregating their results.[PY10, L⁺17].

2 Data

2.1 The WORC Database

The imaging data used in this study originates from the WORC database, a publicly available collection of CT and MR scans with corresponding segmentations and clinical labels assembled specifically for radiomics research on soft tissue tumors [S⁺21, ?]. The database comprises 930 patients from six independent radiomics studies conducted at the Erasmus MC, making it the largest publicly available resource for soft tissue tumor classification research. Each case includes volumetric imaging data in NIfTI format, expert-delineated binary segmentation masks identifying one or more lesion regions, and clinical labels encoding diagnostic outcomes or tumor subtypes.

The database encompasses six distinct tumor categories, each representing a clinically relevant binary classification task. Gastrointestinal stro-

mal tumors (GIST) comprise 246 cases with the classification objective of distinguishing mutation status or malignancy grade. Lipomatous tumors include 115 cases where the task is to differentiate benign lipomas from liposarcomas, a distinction with significant treatment implications. Desmoid-type fibromatosis contains 203 cases focused on predicting tumor progression or stability. Liver lesions encompass 186 cases addressing the differentiation of hepatocellular carcinoma from other liver masses. Colorectal liver metastases (CRLM) include 77 cases with classification targets related to treatment response or resectability. Melanoma metastases comprise 103 cases where the objective is prediction of immunotherapy response or lesion characterization.

2.2 Why WORC: A Challenging Benchmark

The WORC database served as the research choice because it presents one of the most difficult medical image classification benchmarks which researchers can access freely. The dataset contains specific features which make it challenging to work with, but more importantly, it is closer to real-world clinical tasks where foundation models are not adequately evaluated, thus making it an appropriate environment to assess foundation model performance in realistic medical applications

The classification tasks involve subtle imaging differences that are difficult even for expert radiologists to distinguish without histopathological confirmation. The process of identifying benign lipomas from well-differentiated liposarcomas demands pathologists to identify small details which appear identical between these two tumor types [KM13]. The evaluation of GIST tumor mutation status through imaging data represents the maximum extent which visual characteristics can disclose about molecular tumor properties [S⁺19]. If foundation models can achieve reasonable performance on these ambiguous tasks, they are likely to generalize to a wide range of clinical applications where the imaging signal is more pronounced.

The database shows wide variation in imaging methods and scanner brands and scanning methods which affect the six different tumor types. The different characteristics in this data set represent what clinicians experience during their work because they cannot follow standardized procedures which creates difficulties for machine learning models to learn from protocol-based data. A model that achieves high performance in all six categories of WORC demonstrates the stability required for real-world deployment.

The study contains small tumor category sample sizes, including between 77 and 246 cases for each category. While traditional deep learning approaches require large datasets, pre-trained foundation models are particularly well-suited for this data-scarce scenario, as they leverage knowledge from large-scale pre-training to perform effectively with limited task-specific data. The majority of medical imaging applications encounter this limitation because rare diseases and specialized centers and expensive expert annotation services limit the amount of available labeled data. Methods that require thousands of training examples will fail in this regime, making WORC an ideal benchmark for evaluating data-efficient approaches such as foundation model transfer learning combined with small-data classifiers.

2.3 WORC as a Methodological Achievement

The WORC database serves as a reference dataset which brings an essential methodological improvement to medical image analysis and radiomics research. Prior to its release, publicly available datasets for soft tissue tumor classification were scarce, fragmented, and lacked the standardization necessary for reproducible research. The development of WORC solved multiple deficiencies which existed in the previous research.

The database enables standardized data formatting and preprocessing of all six tumor categories which allows researchers to perform unbiased method evaluations without data preparation inconsistencies. All imaging data is provided in NIFTI format with consistent orientation and voxel spacing conventions, and segmentation masks follow a uniform binary labeling scheme. The process of standardization removes a significant research challenge which has affected radiomics studies because different investigations employed incompatible data formats.

The WORC software framework includes an automated machine learning system for radiomics which uses a systematic method to analyze different workflow configurations [?]. The system at WORC uses an automated process to choose between different feature extraction techniques and normalization methods and feature selection approaches and classification models instead of using manual testing to find the best combination. The framework performs configuration assessments through multiple rounds of cross-validation which maintains appropriate training and testing boundaries to solve the radiomics reproducibility problem caused by test set overfit-

Table 1: Summary of the WORC database tumor categories used in this study.

Category	GIST	Lipo	Desmoid	Liver	CRLM	Melanoma	Total
Cases	246	115	203	186	77	103	930
Modality	CT	MRI	MRI	CT/MRI	CT	CT	—

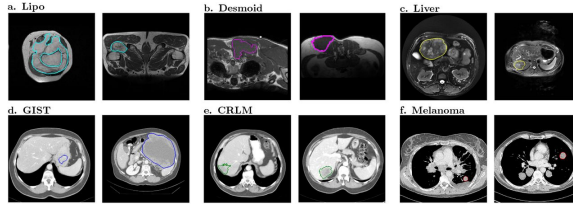


Figure 1: Example CT and MRI slices from the WORC database showing the diversity of tumor appearances and imaging modalities. Top row: GIST and CRLM (CT). Bottom row: lipomatous tumor and desmoid (MRI).

ting and data leakage.

The database and optimization framework exist in public domain which enables researchers to confirm published results through precise duplication and to test new methods by applying them to established benchmarks with established evaluation standards. The authors describe specific cross-validation techniques which allow different research teams to validate their methods by using common training and testing datasets which solves the main challenge for method evaluation. This level of reproducibility infrastructure is rare in medical imaging research and sets a standard for future dataset releases.

2.4 Implications for Foundation Model Evaluation

The combination of task difficulty, data heterogeneity, sample size constraints, and strong baselines makes WORC an ideal proving ground for foundation model approaches to medical image classification. The central hypothesis of this thesis is that pretrained volumetric embeddings from SAM-Med3D, combined with data-efficient tabular classifiers, can achieve competitive performance on these challenging tasks without task-specific training of the image encoder.

The research suggests that foundation models trained on extensive medical imaging data learn representations which successfully apply to subsequent classification tasks that require identifying small visual differences across various imaging methods with restricted available data. The discovery would have significant implications for medical image analysis and AI, as it would decrease the amount of data and computational power needed to develop diagnostic tools from medical images, potentially accelerating the trans-

lation of machine learning research toward clinical applications.

On the contrary, if foundation model transfer learning fails to match optimized radiomics baselines on WORC, it would highlight important limitations of current pretrained models and suggest directions for improving medical imaging foundation models. The results from both scenarios will generate vital information which scientists need to establish the correct operational parameters for foundation model medical image analysis.

3 Methods

The workflow consists of four main stages: data preparation and quality control, feature extraction from volumetric scans using pretrained models, dimensionality reduction and standardization, and classification with tabular foundation models designed for small datasets.

The first stage, data preparation, involves organizing imaging data and metadata from the WORC dataset [S⁺21], a single-center collection comprising multiple tumor categories. The data set contains CT and magnetic resonance volumes that include one or more regions of the lesion that need annotation and their corresponding diagnostic labels. The research team tested different preprocessing methods to find the best combination that would maintain important imaging data while keeping processing time efficient. The evaluation included three methods, which were full-volume processing to maintain complete scan data and lesion size filtering through voxel count and morphological criteria to remove small or poor-quality annotations and lesion-centered region-of-interest cropping with adaptive padding to extract standardized tumor volumes [DGP15].

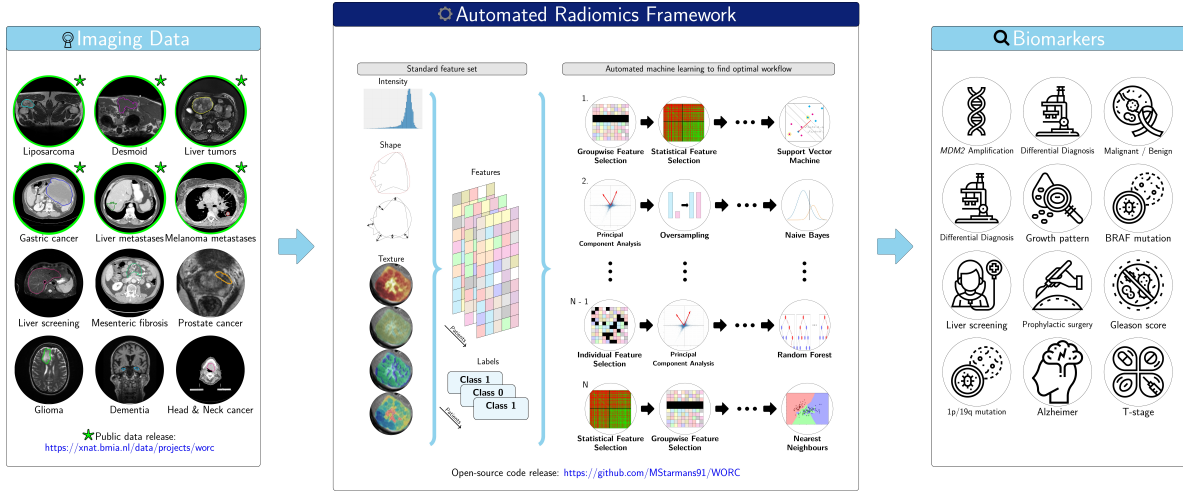


Figure 2: The WORC automated radiomics workflow. The framework systematically evaluates thousands of method combinations to identify optimal pipelines per application, establishing strong baselines for comparison.

The research applied region-of-interest cropping as its main preprocessing technique to meet SAM-Med3D’s fixed input requirement of $128 \times 128 \times 128$ voxels. This approach focuses computational resources on tumor-related features while standardizing all studies to the required input size.

The second stage of processing extracts high-level semantic features from the preprocessed volumes. The analysis uses SAM-Med3D [W⁺23] as its foundation model because it was trained on multiple medical imaging tasks instead of starting from scratch with a volumetric convolutional network. The three-dimensional input of SAM-Med3D gets processed to generate dense embeddings which get distributed across all spatial locations. The model uses these embeddings to create a single feature vector for each study through a percentile pooling layer which combines both appearance and spatial information about the lesion [W⁺23, D⁺21].

The third stage of the process prepares the extracted feature vectors for classification. The features are standardized to zero mean and unit variance, and principal component analysis is applied to reduce dimensionality while retaining the directions of greatest variance [JC16]. The method reduces model overfitting risks while speeding up the training process for future classifiers. For studies with multiple lesions, features are pooled at the patient level according to a predefined aggregation scheme to produce one feature vector per case.

The fourth stage performs binary classification using transformer-based tabular models, specifically TabPFN [HMEH23] and LoCalPFN [T⁺24], which are designed to excel with limited train-

ing data by incorporating strong inductive biases and performing in-context learning. All experiments follow a stratified splitting protocol that preserves class balance in both training and validation sets, and performance is evaluated using accuracy, area under the receiver operating characteristic curve, sensitivity, specificity, and F1 score. Cross-validation or repeated holdout validation is used where sample sizes permit to obtain confidence estimates for each metric.

3.1 Data Preparation and Preprocessing

Several preprocessing strategies were evaluated to determine the optimal trade-off between preserving clinically relevant imaging information and maintaining computational efficiency. Three candidate approaches were considered. Full-volume processing to retain the entire scan without modification before passing it to the encoder; this preserves maximal anatomical context but results in variable input sizes and includes large regions of non-lesion tissue that may introduce noise. Lesion size filtering to apply voxel-count and morphological criteria to exclude annotations that are too small or irregularly shaped to yield reliable features; this improves signal quality but discards a subset of cases and does not address input size variability. Lesion-centered region-of-interest cropping with adaptive padding to extract a fixed-size subvolume centered on the lesion bounding box and pads or crops as needed to produce uniform input dimensions across all studies [DGP15].

SAM-Med3D requires a fixed input size of

128×128×128 voxels, which can be achieved through various preprocessing strategies. Comparative experiments indicated that region-of-interest cropping offered the best approach to meet this requirement for the objectives of this work. Centering the crop on the tumor ensures that the encoder receives input dominated by lesion tissue and its immediate surroundings, reducing the influence of distant anatomy and scanner-specific field-of-view variations. This constraint simplifies batching during feature extraction and ensures that all cases contribute equally to downstream analyses regardless of original scan dimensions. The adaptive padding scheme handles cases where the lesion is located near the edge of the scan by reflecting or zero-padding the boundary, avoiding the need to discard edge cases. Based on these considerations, region-of-interest cropping was adopted as the primary preprocessing method for all subsequent experiments, with lesion size filtering applied as an optional quality-control step when stricter inclusion criteria were desired.

3.2 SAM-Med3D: Foundation Model for Volumetric Medical Imaging

SAM-Med3D is a foundation model designed specifically for three-dimensional medical image segmentation and feature extraction [W⁺23]. The model extends the Segment Anything Model architecture from natural 2D images to volumetric medical data by replacing all 2D operations with their 3D counterparts and adapting the positional encoding to handle three spatial dimensions. The system contains three main components which include a Vision Transformer image encoder that extracts dense embeddings from the input volume and a prompt encoder that processes user-defined guidance through point clicks and bounding boxes and a mask decoder that produces segmentation output. For the purposes of this work, only the image encoder is used to extract feature representations, discarding the prompt encoder and mask decoder components.

The image encoder employs a ViT-B backbone that divides the input volume into non-overlapping 3D patches, projects each patch into a high-dimensional embedding space, and processes the sequence of patch embeddings through multiple transformer layers with self-attention [D⁺21]. The output contains 384 channels which show both local texture details and complete anatomical information at a decreased spatial scale. The dense embeddings need percentile pooling to combine them into single feature vectors which func-

tion as compact representations for classification tasks.

The training data comes from SA-Med3D-140K which is a large collection of medical segmentation data that includes 143,000 3D masks which cover 245 anatomical categories and uses CT and MRI imaging modalities [W⁺23]. This scale of pretraining far exceeds what is typically available for individual tumor classification tasks and enables the model to learn generalizable representations of medical anatomy and pathology. Second, the model achieves efficient promptable segmentation that requires 10 to 100 times fewer prompt points than previous methods to produce satisfactory 3D segmentation results, indicating that the learned representations capture clinically meaningful structure. Third, the SAM-Med3D-turbo variant used in this work was further fine-tuned on 44 additional datasets to improve cross-domain performance, making it particularly robust to variations in scanner protocols, image quality, and anatomical regions [W⁺23].

The choice of SAM-Med3D as the feature extraction backbone for this work is motivated by several considerations that align with the constraints and objectives of the classification task. The primary advantage is the ability to leverage rich pretrained representations without requiring large amounts of labeled tumor classification data. The process of training a volumetric encoder with similar characteristics from start would produce excessive model specialization because the SAM-Med3D encoder starts with a pre-trained model which achieves good results in different body areas and disease types. The model contains specific design elements which make it suitable for medical imaging because it handles the typical intensity ranges and noise patterns and anatomical features found in CT and MRI scans. The model differs from general-purpose vision models because it natively processes 3D volumetric data rather than analyzing individual slices, and was specifically trained on medical imaging rather than natural images. [PY10, L⁺23].

Another practical consideration is computational efficiency. Pre-trained foundation models, including SAM-Med3D, can be used in a frozen configuration, meaning that feature extraction requires only a single forward pass through the network without gradient computation. This characteristic of foundation models dramatically reduces memory requirements and processing time compared to end-to-end training of a volumetric classification network. The extracted features become available for storage so they can be used in various classification experiments which allows re-

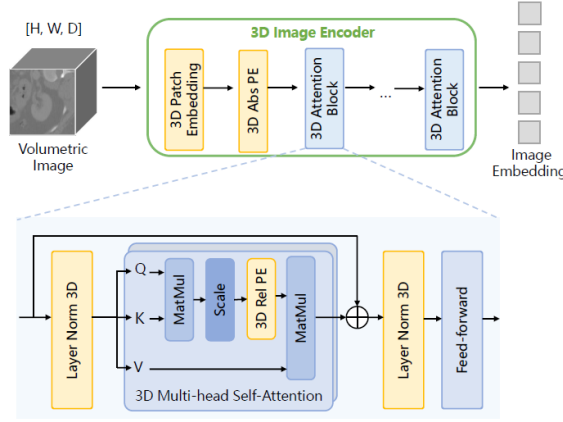


Figure 3: Architecture of SAM-Med3D. The model consists of a 3D Vision Transformer image encoder, a prompt encoder for interactive guidance, and a mask decoder. For feature extraction, only the image encoder is used.

searchers to test different model choices and hyperparameters and preprocessing methods at a fast pace without needing to perform the costly encoding process again. The experimental design of this work depends on separating feature extraction from classifier training because this approach enables researchers to evaluate transfer learning against end-to-end methods through controlled experiments.

3.3 Feature Standardization and Dimensionality Reduction

The third stage prepares the extracted feature vectors for classification through standardization and dimensionality reduction. Study-level embeddings produced by SAM-Med3D’s encoder have 384 channels and by default are pooled with percentile pooling (capturing five quantiles: 10th, 25th, 50th, 75th, 90th) to yield a 1,920-dimensional vector per case. The vectors undergo standardization to achieve zero mean and unit variance using training partition statistics, which prevent information leakage. Following standardization, PCA reduces the dimensionality while retaining the most informative variance. The standardization process ensures all feature dimensions contribute equally, preventing large numerical values from dominating the subsequent learning process.

The method of principal component analysis serves as an optional procedure which researchers should use when their pooling or aggregation approach generates output data that exceeds typical dimensions (this occurs when researchers combine different lesion embeddings through concatenation or when they replace average pooling

with attention schemes or concatenation methods). When triggered, PCA is fitted on the standardized training features and the number of retained components is chosen as the minimum of a preset cap (default 500) and the numerical rank of the training feature matrix [JC16]. The default 384-dimensional average-pooled vectors disable PCA but the method becomes effective when different pooling techniques increase the data dimensions. The PCA transform which was trained on the data gets applied to both training and validation features to produce the final reduced representations.

Spatial pooling methods aggregate features within each lesion volume to create a fixed-size embedding. Three aggregation strategies are available—average pooling, multiscale pooling, and percentile pooling. Only average pooling produces a compact enough representation to avoid PCA dimensionality reduction; the other strategies generate higher-dimensional vectors that require PCA before classification. The output of this stage is a set of normalized, optionally reduced feature vectors (one per case), with associated labels and identifiers, which serve as the input to the classification models.

3.4 TabPFN: Transformer-Based Classification for Small Tabular Datasets

The TabPFN (Tabular Prior-data Fitted Network) functions as a transformer-based classifier which delivers optimal results on small tabular classification problems at fast speeds [HMEH23]. The single forward pass of TabPFN enables classi-

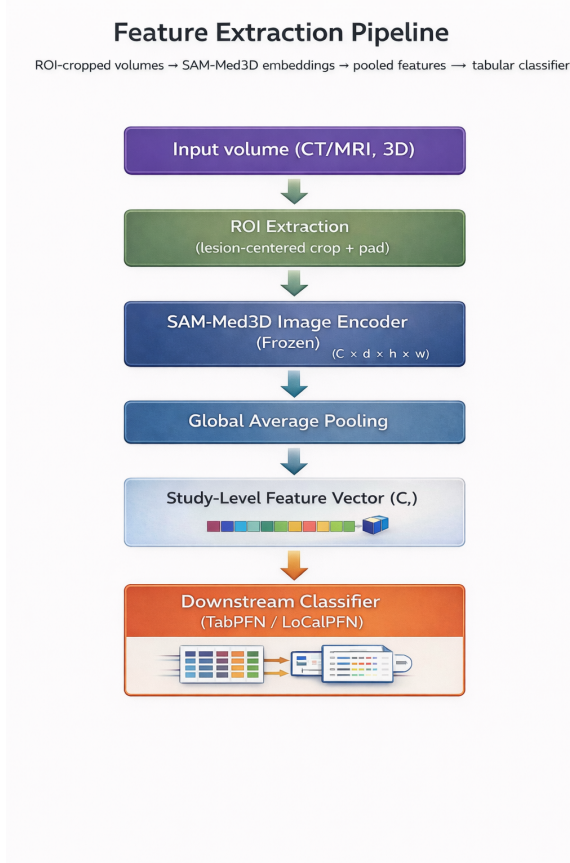


Figure 4: Feature extraction pipeline using SAM-Med3D. Input volumes are preprocessed with ROI cropping, passed through the frozen image encoder, and aggregated via percentile pooling to produce feature vectors used for downstream classification.

fication because it uses in-context learning to make predictions based on training examples provided during inference without any parameter updates. The model in in-context learning uses a single forward pass to process labeled examples which enables it to learn new tasks without needing parameter updates or gradient-based training. TabPFN applies this approach to tabular classification by treating supervised learning as a sequence modeling problem: the model is pre-trained on synthetic tabular datasets and learns to predict labels by jointly attending to training examples (provided as context) and test samples within a transformer architecture. The inference process of TabPFN uses the complete training data and test sample in a single forward computation to achieve classification results without model retraining which benefits small datasets that need deep learning models to be retrained at high cost. This approach fundamentally changes how tabular classification is performed: instead of fitting a model to the data,

the data is provided as context to a pretrained model that has learned to generalize across a vast distribution of synthetic classification tasks.

The architecture of TabPFN is based on a transformer encoder that processes both training and test samples simultaneously [HMEH23, D⁺21]. During inference, the training set features and labels are concatenated with the test set features and passed through the transformer, which attends to all samples jointly. The output for each test sample is a probability distribution over the target classes, computed by attending to the training examples and implicitly learning decision boundaries in feature space. The design allows TabPFN to handle datasets with 10000 training examples and 500 features which it can process at a rate of one second on modern GPUs while outperforming other methods that require hyperparameter optimization and cross-validation.

The pretraining procedure of TabPFN is central to its effectiveness. The model was trained on millions of synthetic classification datasets generated from a carefully designed prior distribution that captures the structural properties of real-world tabular data [HMEH23]. This prior, called the Prior-data Fitted Network prior, samples datasets with varying numbers of features, class distributions, feature correlations, and non-linear decision boundaries. By training on this diverse distribution of tasks, TabPFN learns a general-purpose classification algorithm that can be applied to new datasets without task-specific training. The pretraining objective is to minimize the cross-entropy loss on held-out samples from each synthetic dataset, effectively teaching the model to perform Bayesian posterior inference over the space of possible classifiers given the observed training data.

The TabPFN approach brings new research methods to the field because of its several key advantages. The method removes the requirement for hyperparameter optimization which represents the most prolonged step when machine learning models need to be applied to fresh data sets. The performance of traditional methods including gradient boosting and random forests and support vector machines depends heavily on the selection of learning rates and tree depths and regularization parameters and other critical settings. TabPFN produces excellent results through a single pretrained model which operates without requiring any specific dataset modifications to simplify machine learning operations. The in-context learning paradigm functions as an implicit ensemble system because the transformer model processes all training examples to generate predic-

tions which combine all dataset information for detecting intricate feature relationships without needing separate feature development. The results from TabPFN show that transformers achieve performance levels which match or surpass those of gradient boosting methods that received extensive optimization when working with small tabular datasets thus contradicting the established belief that tree-based models perform best in this field [HMEH23, GOV22].

The innovation extends to the theoretical foundations of the approach. TabPFN can be interpreted as performing approximate Bayesian inference over the posterior predictive distribution of labels given the training data [HMEH23, MHA⁺23]. Under this interpretation, the pre-trained transformer has learned to approximate the integral over all possible classifiers weighted by their posterior probability given the observed data and the prior. The method provides a statistical framework which explains why TabPFN achieves good results because it avoids selecting one particular hypothesis while it considers all possible models which match the observed data. The synthetic pretraining distribution contains structural information about real-world tabular data which resembles how Gaussian process priors use smoothness assumptions to function space.

The selection of TabPFN as our main classification model stems from multiple operational and research-based factors which match the requirements of soft tissue tumor identification. The main benefit of this approach emerges when working with limited datasets. Medical imaging research which examines unusual tumors usually works with participant numbers that reach into the hundreds but do not exceed thousands. The current deep learning classifiers fail to perform well in this environment because they develop overfitting problems yet TabPFN was created and trained for working with datasets which contain less than 10000 examples. The SAM-Med3D embeddings which extract features from tumor volumes generate vectors of average size which TabPFN can process after dimensionality reduction thus creating a suitable match between the two methods.

Another practical consideration is reproducibility and experimental throughput. Because TabPFN requires no training or hyperparameter selection, results are deterministic given the same input data, and experiments can be iterated rapidly. This is particularly valuable when comparing multiple preprocessing strategies, feature extraction configurations, and dataset combinations, as is the case in this thesis. A single exper-

imental configuration can be evaluated in seconds rather than the hours required for cross-validated hyperparameter search with traditional methods. This efficiency enabled systematic exploration of ROI cropping parameters, lesion filtering thresholds, and dimensionality reduction settings that would have been impractical with slower classifiers.

The TabPFN integration process occurs after the feature preparation stage in the classification pipeline. After SAM-Med3D embeddings are extracted and pooled to study-level vectors, the features are standardized to zero mean and unit variance using statistics computed on the training set. The analysis uses Principal component analysis to decrease data dimensions into 500 components which fulfills TabPFN feature restrictions and protects against high-dimensional space overfitting with insufficient sample data. The reduced features are passed to TabPFN along with binary labels, and class probabilities are obtained in a single forward pass. The TabPFN system enables users to make ensemble predictions through different attention configurations which produce combined results for better stability but our initial tests revealed no performance gain for the evaluated datasets.

The model has a hard limit of 10000 training samples, which is not a constraint for the datasets in this study but would require alternative approaches for larger cohorts. Additionally, TabPFN was pretrained on synthetic data with specific distributional assumptions that may not perfectly match all real-world datasets. The thesis evaluates how different classifier selections affect result outcomes through a comparison with LoCalPFN which uses TabPFN to perform local neighborhood retrieval for better results on datasets that show better local than global patterns [T⁺24]. Native 3D classification heads trained end-to-end provide an additional baseline that does not rely on the TabPFN architecture.

3.5 LoCalPFN: Local Context Adaptation

While TabPFN achieves strong performance by conditioning on all training examples simultaneously, this global attention mechanism may not be optimal when local structure in feature space is more informative than global patterns [HMEH23]. The system LoCalPFN solves this problem through its combination of k-nearest-neighbor retrieval with TabPFN inference which gives each test sample access to a relevant set of training examples for context [T⁺24].

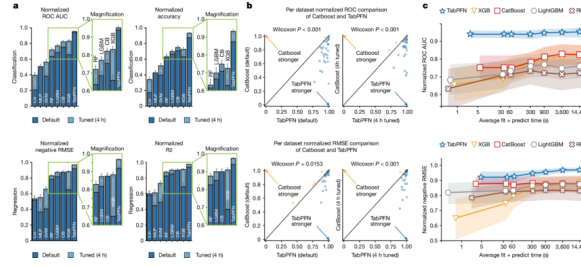


Figure 5: Performance comparison of TabPFN against traditional machine learning methods. TabPFN achieves competitive accuracy while requiring no hyperparameter tuning and completing inference in significantly less time.

The core idea is straightforward: rather than passing the entire training set to TabPFN, the method first retrieves the k most similar training samples to each test query using Euclidean distance in the standardized feature space [CH67]. The TabPFN system conducts inference operations based on the local neighborhood which serves as its context to generate predictions that focus on the most important examples instead of distant or unrelated cases. The neighborhood size k is set automatically as the minimum of 1000 and ten times the square root of the training set size, balancing local specificity against having sufficient context for reliable inference [T+24]. In our datasets (77-246 samples per category), this formula yields k values ranging from approximately 88 to 157 neighbors, ensuring that LoCalPFN uses local context while the 1000-sample ceiling prevents excessive computational cost in larger datasets.

The local method provides better results when feature space boundaries differ between different areas and when the data includes multiple distinct population groups [AMS97]. Medical imaging applications require different tumor subtypes to stay in separate areas of the embedding space because these areas have their own specific decision boundaries which global models fail to detect. The LoCalPFN model restricts its attention to nearby examples which enables it to make predictions based on local class distributions and feature relationships according to [T+24, L+17].

The implementation in this work extends the retrieval-based approach with an optional lightweight adapter layer trained on TabPFN logits [HGJ+19]. The adapter system allows the model to find particular modifications for base TabPFN output predictions through neighborhood information which enhances local prediction results without modifying the protected TabPFN weights. The adapter training uses a small number of epochs with modest learning rates to avoid overfitting while capturing any systematic biases in the local predictions.

Experimental comparison between TabPFN and LoCalPFN across the WORC tumor categories tests whether local or global context provides better discriminative power for these classification tasks [S+21]. The two methods use identical data processing steps which include data preparation and both extract features and reduce dimensions but they differ in their prediction algorithms.

References

- [AMS97] Christopher G. Atkeson, Andrew W. Moore, and Stefan Schaal. Locally weighted learning. *Artificial Intelligence Review*, 11(1–5):11–73, 1997.
- [CH67] Thomas Cover and Peter Hart. Nearest neighbor pattern classification. *IEEE Transactions on Information Theory*, 13(1):21–27, 1967.
- [D⁺21] Alexey Dosovitskiy et al. An image is worth 16x16 words: Transformers for image recognition at scale. *International Conference on Learning Representations*, 2021.
- [DGP15] Ivo Despotović, Bart Goossens, and Wilfried Philips. Preprocessing in medical image analysis: A review. *Medical Image Analysis*, 26(1):79–107, 2015.
- [Fle18] Christopher D. M. Fletcher. Soft tissue tumors: An update on diagnosis and treatment. *Surgical Pathology Clinics*, 11(3):579–589, 2018.
- [GOV22] Léo Grinsztajn, Edouard Oyallon, and Gaël Varoquaux. Why do tree-based models still outperform deep learning on tabular data? *Advances in Neural Information Processing Systems*, 35:507–520, 2022.

- [HGJ⁺19] Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin De Laroussillhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. Parameter-efficient transfer learning for nlp. *International Conference on Machine Learning*, pages 2790–2799, 2019.
- [HMEH23] Noah Hollmann, Samuel Müller, Katharina Eggenberger, and Frank Hutter. Tabpfn: A transformer that solves small tabular classification problems in a second. *International Conference on Learning Representations*, 2023. Available in papers/tab.pdf.
- [JC16] Ian T. Jolliffe and Jorge Cadima. Principal component analysis. *Philosophical Transactions of the Royal Society A*, 374(2065):20150202, 2016.
- [KM13] Mark J. Kransdorf and Mark D. Murphy. Imaging of soft tissue tumors. *Seminars in Musculoskeletal Radiology*, 17(2):162–173, 2013.
- [L⁺17] Geert Litjens et al. Deep learning in medical image analysis. *Medical Image Analysis*, 42:60–88, 2017.
- [L⁺23] Xiaoxiao Liu et al. Foundation models for medical imaging: A survey. *Medical Image Analysis*, 87:102834, 2023.
- [MHA⁺23] Samuel Müller, Noah Hollmann, Sebastian Pineda Arango, Josif Grabocka, and Frank Hutter. Prior-data fitted networks are effective bayesian inference models. *Advances in Neural Information Processing Systems*, 36, 2023.
- [PY10] Sinno Jialin Pan and Qiang Yang. A survey on transfer learning. *IEEE Transactions on Knowledge and Data Engineering*, 22(10):1345–1359, 2010.
- [S⁺19] Benoit Schmauch et al. Radiomics and radiogenomics in gastrointestinal stromal tumors. *Abdominal Radiology*, 44:1986–1996, 2019.
- [S⁺21] Martijn P. A. Starmans et al. Worc: A comprehensive open-source platform for radiomics research. *Journal of Medical Imaging*, 8(3):031101, 2021. Available in papers/worc.pdf.
- [T⁺24] Samuel Thomas et al. Localpfn: Locally calibrated prior-fitted networks for tabular data. *arXiv preprint*, 2024. Available in papers/local.pdf.
- [W⁺23] Haoyu Wang et al. Sam-med3d: Towards foundation models for medical volumetric segmentation. *arXiv preprint arXiv:2310.15161*, 2023. Available in papers/sammed3d.pdf.